

The Mirage of Model Editing: Revising Evaluation in the **WILD**

Wanli Yang, Fei Sun, Jiajun Tan, Xinyu Ma, Qi Cao, Dawei Yin, Huawei Shen, Xueqi Cheng

State Key Laboratory of AI Safety, Institute of Computing Technology, CAS

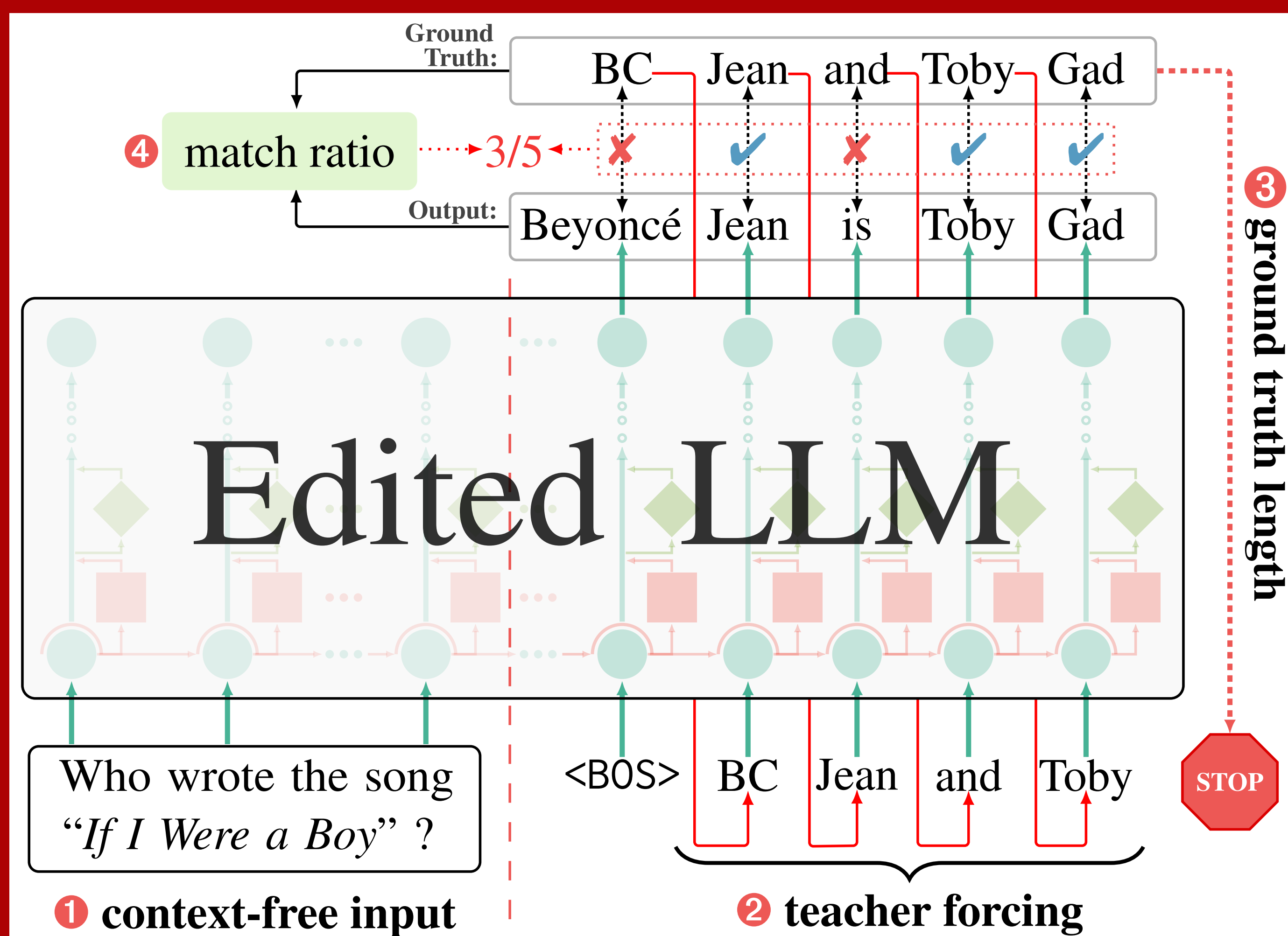
University of Chinese Academy of Sciences Baidu Inc.

✉ yangwanli24z@ict.ac.cn sunfei@ict.ac.cn

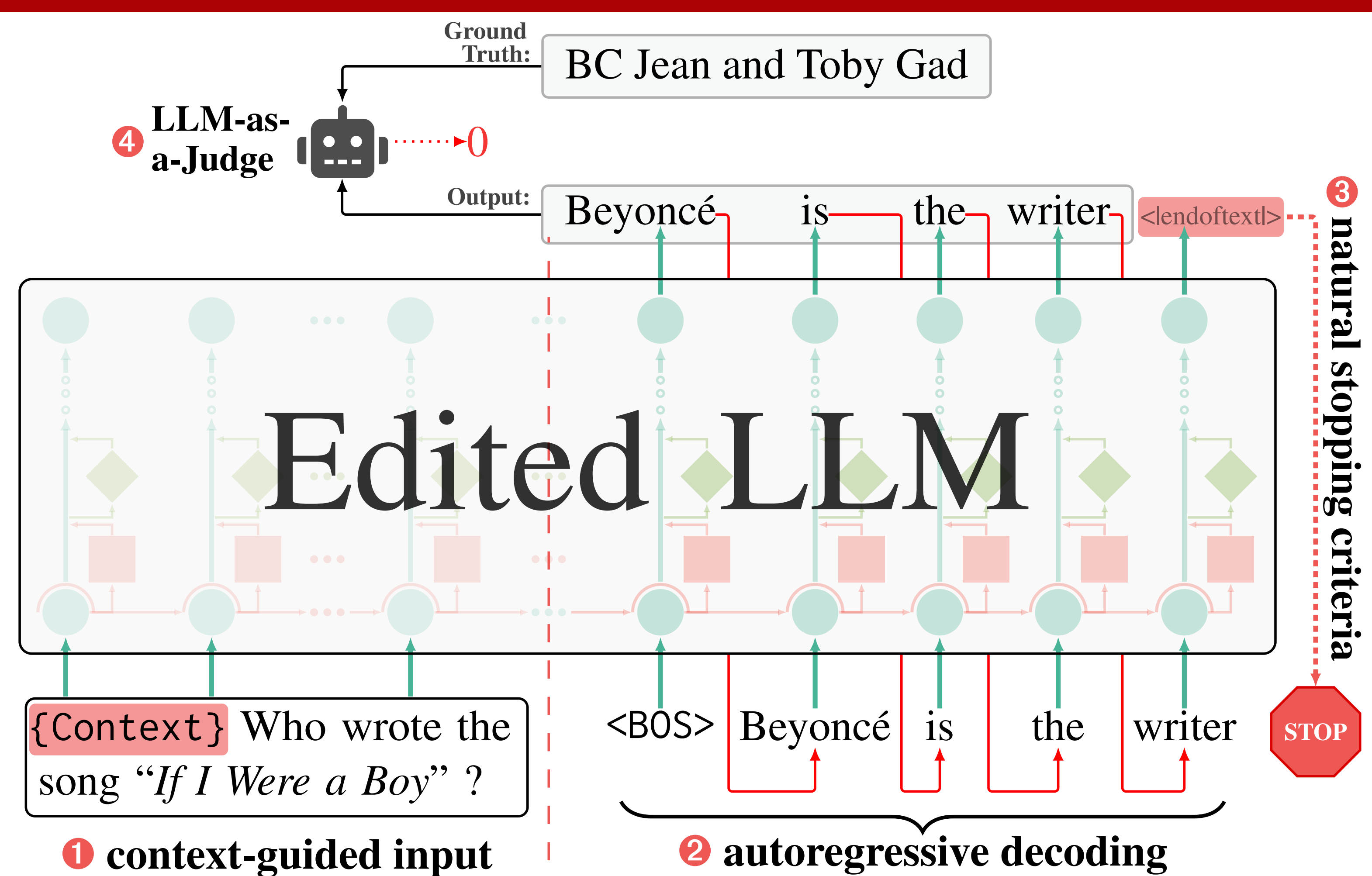
ACL 2025
VIENNA
JULY 27 - AUGUST 1



Stop Using *Teacher Forcing* to Evaluate Model Editing!



(a) synthetic evaluation framework



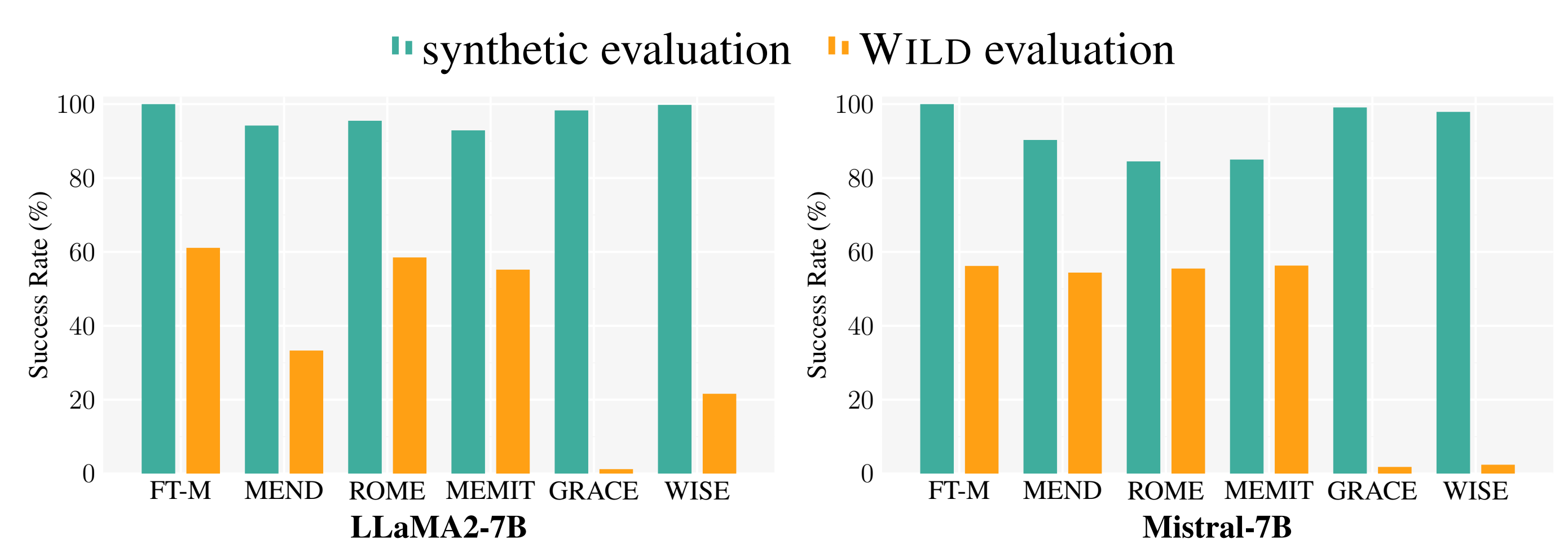
(b) WILD evaluation framework

Comparison of Two Evaluations

Module	synthetic	WILD
Input	context-free	context-guided
Gen. Strategy	teacher forcing	autoregressive decoding
Output Trunc.	ground truth length	natural stopping criteria
Metric	match ratio	LLM-as-a-Judge / EM

- Previous **synthetic** evaluation relies on oversimplified and unrealistic settings, most notably **teacher forcing**, which **leaks** both the **content** and **length** of **ground truth**.
- We propose **WILD** (Without Intervention, Live Decoding) to align with real-world inference conditions.

Evaluate Editing in the WILD



- Editing success rate **drops sharply** from synthetic evaluation to WILD evaluation for all techniques!
- Which modules in synthetic evaluation **contribute most** to performance **overestimation**?

Teacher Forcing Generation

Generation Strategy	FT-M	ROME	MEMIT	GRACE	WISE
① context-free, ③ ground truth length, ④ match ratio					
teacher forcing	1.000	0.985	0.965	0.998	0.908
autoregressive decoding	1.000	0.967	0.929	0.996	0.765
① context-guided, ③ ground truth length, ④ match ratio					
teacher forcing	0.937	0.930	0.907	0.412	0.838
autoregressive decoding	0.800	0.851	0.786	0.036	0.592

- Switching from teacher forcing to autoregressive decoding **consistently leads to performance degradation**!
- Underlying reason is that teacher forcing prevents error propagation by **feeding ground truth** tokens as input.

Ground Truth Length Truncation

Irrational truncation **masks subsequent errors** in real-world generation.

Truncation Strategy	FT-M	ROME	MEMIT	GRACE	WISE
① context-free, ② autoregressive decoding, ③ LLM-as-a-Judge					
ground truth length	1.000	0.954	0.886	0.992	0.700
natural stop criteria	0.202	0.478	0.461	0.301	0.046
① context-guided, ② autoregressive decoding, ③ LLM-as-a-Judge					
ground truth length	0.751	0.783	0.704	0.003	0.482
natural stop criteria	0.528	0.556	0.529	0.000	0.108

Irrelevant Information	
Input Prompt	Who was the first lady nominated member of the Rajya Sabha?
Target Answer	Mary Kom
Natural Stop	Mary Kom is the first woman boxer to qualify for the Olympics
Incorrect Information	
Input Prompt	When does April Fools' Day end at noon?
Target Answer	April 1st
Natural Stop	April 1st ends at noon on April 2nd

Model editing should be made more than just a few tokens deep!